TAYLOR FLAAT

Senior Research Scientist · Immunology & Molecular Assay Development · Preclinical Models · Method Development & Qualification

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PROFESSIONAL SUMMARY

Research scientist with ten years across academia, venture-backed biotech, CRO, and large pharma, specializing in standing up preclinical methods where none previously existed. Built and qualified cell-based, immunological, and molecular assays (multicolor flow cytometry and FACS, ELISA, Western blot, live-cell imaging, qPCR/RT-qPCR) across 13+ mammalian cell lines and primary skin and oral mucosa tissue. Drove an IND-enabling VSV-vectored vaccine from construct design to pre-IND submission in five months, holding the critical path across internal production, an external in vivo partner, and a regulatory workstream. GLP/GCP-trained BSL-2 operator, SOP and test-method author, peer-reviewed author (Viruses 2024), AACR presenter, and contributor to patented technology.

EDUCATION & CERTIFICATIONS

M.S. Biomedical Sciences · Auburn University College of Veterinary Medicine 2020

B.S. Biomedical Sciences · University of South Florida · Minor: Biomedical Physics 2016

Certifications BSL-2 laboratory operations · GLP / GCP Training · Google AI Professional Certificate (2026)

CORE COMPETENCIES

Immunology Cell-based immune & potency assays · CellTiter-Glo viability · Multicolor flow cytometry immune profiling · ELISA (cytokine & protein quantification) · Western blot & SDS-PAGE · Immunofluorescence · Mast-cell progenitors & IgE-receptor-bearing effector lineages ($CD117^+$ $Fc\varepsilon RI^+$ $CD90^-$) · Immunotherapy assays (oncology, autoimmune)

Cell & Tissue Mammalian cell & tissue culture (13+ lines) · Primary cells from ex vivo tissue: punch biopsy → enzymatic digestion → culture · Skin & oral mucosa; epidermal/dermal separation · PBMC isolation & blood-derived assays · 3D spheroid & organoid culture · Multi-cell-type co-culture · Cell factories & scaled adherent culture · Single-cell isolation & rare-cell sorting; MACS depletion / enrichment; FACS · Aseptic technique

Assays & PCR qPCR / RT-qPCR / dPCR-ready workflows · Assay development, optimization & qualification · Viral titer & genomic-copy quantification (10^3-10^{12}) · TCID50 & plaque assay (infectious titer) · Primer design & amplification strategy · Gel electrophoresis · Sanger & NGS construct verification

Gene Editing CRISPR/Cas9 gene editing · AAV, lentivirus, oncolytic VSV · Gibson, Golden Gate & seamless cloning · Site-directed mutagenesis · Bacterial transformation & culture · Construct design → rescue → production → purification → titration · Full-genome assembly (11,161 bp VSV) · Glycoprotein retargeting & aptazyme switches · Ultrafiltration/diafiltration (UF/DF)

Genomics & NGS Illumina, Oxford Nanopore, PacBio · DNA/RNA library prep & QC · Multiplexed / barcoded library workflows · Direct RNA-seq (SQK-RNA004 chemistry) · Bisulfite conversion, methylation & epigenomics · Base-calling, alignment & variant ID · Single-cell sequencing pipelines · FFPE extraction & prep

Instruments Incucyte live-cell imaging · Qubit, TapeStation, qPCR (QC) · BioMek, Dragonfly, FAST (automation) · HPLC · Confocal & fluorescence microscopy · Geneious, IGV & Galaxy (sequence analysis) · GraphPad Prism; ELN systems · SOP & test-method authoring

PROFESSIONAL EXPERIENCE

Scientist I

Sept 2024 – Mar 2026

Azenta Life Sciences · CRO for next-generation sequencing applications

- Managed a parallel portfolio of client sequencing programs, each with distinct scope, QC gates, and fixed delivery dates; triaged compromised inputs (degraded FFPE) against schedule risk and delivered sequenceable libraries without moving the date.
- Developed, optimized, and qualified custom DNA/RNA NGS library-prep workflows across Illumina, Oxford Nanopore, and PacBio, improving throughput and reproducibility across high-volume projects under standardized protocols, including CRISPR-validation, targeted-genomics, and whole-genome applications.
- Executed molecular cloning and primer design alongside the NGS workload: plasmid preparation and purification, bacterial transformation and culture, colony screening, and amplification strategy for targeted-genomics panels, with constructs confirmed by Sanger sequencing and analyzed in Geneious.
- Planned and executed concurrent DNA-prep workflows independently with limited oversight, following vendor kit protocols and adapting them for degraded and low-input material; worked issues to root cause and communicated corrective actions to project teams, keeping records complete and traceable.
- Maintained standardized QC via Qubit, TapeStation automated electrophoresis, and qPCR, performing data review and verification on high-volume sample sets; managed equipment, calibration, and lab inventory for the automation fleet (BioMek, Dragonfly, FAST), and mentored junior scientists and interns on library-prep methods, automation, and molecular-biology best practices.

Scientist II, Associate (Contract)

Feb 2024 – Sept 2024

Merck via On-Board Companies · Rahway, NJ · Sanger and next-generation sequencing, library preparation

- Executed advanced molecular cloning (Gibson Assembly, Golden Gate Assembly, site-directed mutagenesis) and DNA library building supporting R&D in genetic engineering and synthetic biology within a large-pharma environment.
- Performed Nanopore, Ion Torrent, and Sanger sequencing for genetic validation, genome assemblies, and construct-integrity confirmation; carried out DNA/RNA extraction, plasmid purification, transformation, electroporation, and colony PCR for gene-of-interest investigation.
- Coordinated deliverables across internal and external teams from experimental design through data delivery, translating results into written reports and recommendations that informed project decisions.

Research Associate II Dec 2021 – Oct 2023

Humane Genomics · New York, NY · Synthetic-virology startup engineering viruses for precision cancer therapy

- Led the VSV-vectored COVID-19 vaccine candidate from construct design through in vivo safety data in five months: displayed SARS-CoV-2 spike on a VSV backbone, confirmed ACE2-receptor binding by live-cell imaging, scaled production and purification 100-1000x for animal work, and supported a rat safety study returning clean CBC and clinical chemistry.
- Held that program's critical path across internal design and production, an external in vivo partner at Auburn's Scott-Ritchey Research Center, and a regulatory workstream closing with a pre-IND FDA submission; reported to founders and stakeholders on a written cadence.
- Co-led the synthetic VSV platform program, taking an 11,161 bp genome from modular fragment design through assembly, rescue, and titration, then demonstrating design freedom by swapping a foreign glycoprotein and rearranging gene order; sequenced hundreds of engineered virions for construct verification and closed with a peer-reviewed paper in Viruses (2024).
- Engineered retargeted oncolytic VSV with aptazyme replication switches for GPC3-positive liver cancer, demonstrating selective lysis at low MOI in vitro and in vivo tolerability of 10^7 PFU doses (presented at AACR 2023).
- Built the company's quantification and sequencing capability from no prior infrastructure: delivered a two-step RT-qPCR method spanning 10^3-10^{12} genome copies after one-step chemistry failed on low-yield material, plus Nanopore sequencing and assembly workflows for construct verification, with test methods and SOPs that made each auditable and repeatable; directly analogous to VCN, genomic-titer, and infectious-titer QC methods.
- Redesigned viral screening into a 24-plex, 24-hour Nanopore pipeline (viral RNA isolation → cDNA → PCR → Rapid Barcoding Kit → assembly), turning a serial multi-day task into a parallel screen, then authored the SOPs and trained the team so it became standard practice; separately compressed plasmid QC sequencing to a two-hour miniprep-to-analysis workflow.
- Built multicolor flow cytometry / FACS and cell-based assays to characterize transduction efficiency, target engagement, viability, and immune populations across cancer and immune-cell models, supporting immunotherapy studies in oncology and autoimmune disease.
- Established 3D spheroid and organoid cultures and developed multi-cell-type co-cultures pairing tumor cells with immune-cell populations; read out viability, proliferation, and drug response by confocal microscopy, Incucyte kinetic imaging, CellTiter-Glo, ELISA, Western blot, and immunofluorescence.
- Designed, built, and validated engineered viral vectors (AAV, lentivirus, oncolytic VSV) using Gibson and Golden Gate Assembly, site-directed mutagenesis, and CRISPR/Cas9; scaled upstream production in multi-layer cell factories, purified stocks by UF/DF and density gradient, and ran ELISA and Western blot for titer and purity QC.
- Ran the lab's experimental workflow across 13 active mammalian cell lines concurrently; trained new researchers on assay design and execution.

Research Associate I → Research Technician Nov 2019 – Dec 2021

Humane Genomics · New York, NY · Founding bench scientist

- Founding laboratory scientist, one of three people at the company: planned and stood up the entire bench and QC operation, covering experimental workflow design, instrument and consumable resource planning, and the foundational SOP set that let the company scale past its founding team.
- Contributed Nanopore sequencing and viral-genome assembly for construct verification across iterative design cycles.

ACADEMIC RESEARCH

Graduate Research Assistant 2016 – 2020

Auburn University College of Veterinary Medicine · Pathobiology · Dr. Bruce Smith & Dr. Tatiana Samoylova

- Isolated circulating mast-cell progenitors of the CD117$^+$ FcεRI$^+$ CD90$^-$ phenotype, the FcεRI-bearing effector lineage of IgE-mediated allergic response, developing rare-cell isolation protocols (MACS CD90$^+$ depletion / CD117$^+$ enrichment → FACS) with viability gating and fluorescence-minus-one controls.
- Delivered the first published method for isolating melanocytes from canine skin and oral mucosa via differential adhesion in phorbol-ester-free Melanocyte Growth Medium M2, a protocol since independently reproduced by other labs.
- Cultured primary cells from ex vivo tissue: punch biopsy → Dispase Grade II digestion → epidermal/dermal separation → TrypLE dispersal → selective-medium expansion of melanocytes and keratinocytes, delivering single-cell-sequencing-grade material for precision oncology.
- Supported a companion-animal contraceptive program using Gibson Assembly, site-directed mutagenesis, PCR, and RT-PCR; authored a Master's thesis at Auburn's Scott-Ritchey Research Center.

SELECTED PUBLICATIONS & PRESENTATIONS

Leveraging Synthetic Virology for the Rapid Engineering of Vesicular Stomatitis Virus (VSV)
Moles CM, Basu R, Weijmarshausen P, Ho B, Farhat M, Flaat T, Smith BF · Viruses 2024, 16(10):1641 · doi:10.3390/v16101641 · Peer-reviewed, open access

Oncolytic VSV with Retargeted Glycoprotein and Genetic On-Switch for Precise Targeting of Liver Cancer
Moles CM, Basu R, Flaat T, Ho B, Farhat M, Weijmarshausen P, Smith BF, Whitt M · Poster, AACR Annual Meeting, April 2023

Methods for Isolating Canine Mast Cell Progenitors, Melanocytes, and Keratinocytes
Flaat TD · Master's Thesis, Auburn University College of Veterinary Medicine, April 2020 · etd.auburn.edu/handle/10415/7140

Patents: contributor to patented synthetic-virology technology (Humane Genomics). Details available on request.

ADDITIONAL EXPERIENCE

Co-Owner & Technical Lead · UntriviallyJess 2024 – Present

Jersey City, NJ · untriviallyjess.com

- Co-founded and co-own a Jersey City live-events company; built untriviallyjess.com end to end (vanilla HTML/CSS/JS; GitHub → Netlify → Cloudflare) with a GA4-instrumented booking flow, a serverless client and venue portal, and a Google Cloud Run leaderboard app scoring six season-long metrics.
- Ran the operating side end to end: venue contracts, host recruiting and scheduling, numbered invoicing, host payouts gated on invoice payment, and a self-built desktop payroll and accounting app (invoices, expenses, W-2 and 1099 tracking, mileage, estimated tax, Schedule C).